

MULTI-FEATURE GENERATIVE WARDROBE & STYLE PROMPT ENGINE

Technical Report

1. Physical Architecture Module

Organic Curvilinear Closet System

- **Design:** Parametric design principles are employed to generate fluid, organic, and curvilinear forms for the wardrobe system. This allows for adaptive spatial configurations and aesthetic integration into diverse interior environments.
- **Material Specifications:**
 - Primary Structure: High-density, sustainably sourced composite wood with a matte, fingerprint-resistant finish.
 - Accents: Polished brushed aluminum or sustainably sourced bamboo for contrast and durability.
 - Interior Linings: Soft-touch, anti-static fabric in a neutral tone (e.g., light grey or ecru).
- **Dimensional Details:**
 - Customizable to client space, with standard module dimensions optimized for efficient storage and accessibility.
 - Depth: 600mm - 700mm for ample hanging and shelving space.
 - Height: Variable, designed to integrate with ceiling heights.
 - Width: Modular, allowing for expansion and custom configurations.

2. Digital Outfit Engine

Fabric Simulation & Rendering

- **Hydrophobic Fabric Simulation:** Advanced physics-based simulation accurately models the behavior of hydrophobic materials. This includes droplet behavior, surface tension effects, and water repellency, ensuring realistic visual representation.
- **Waterproof Textile Rendering:** Utilizes PBR (Physically Based Rendering) techniques to capture the unique light interaction properties of waterproof textiles. Emphasis on sheen, texture, and the subtle variations that indicate water resistance.
- **Weather-Resistant Layering:** The engine supports the simulation and rendering of multiple weather-resistant layers. This includes the interplay of different materials (e.g., shells, insulation,

liners) and their combined visual and functional properties under various simulated weather conditions (rain, wind, light snow).

3. Style & Biometric Analysis

- **Body Scanning:** Integration with high-resolution 3D body scanning technology to capture precise anthropometric data.
- **Silhouette Optimization:** Algorithms analyze body scan data to determine flattering silhouettes and optimal garment fits.
- **Color Harmony:** Utilizes color theory and biometric feedback (e.g., skin tone, eye color) to suggest complementary and harmonious color palettes for outfits.
- **Facial Geometry Mapping:** Advanced facial recognition and mapping to understand individual facial structure. This data can be used to inform accessory choices (e.g., eyewear, hats) and to provide a more holistic style recommendation.
- **Personalized Style Profiles:** Combines biometric data, user preferences, and historical fashion trends to build comprehensive and evolving style profiles for each user.

4. Generative Image Output

Photorealistic Rendering & Quality Assurance

- **Photorealistic Rendering System:** Employs cutting-edge rendering pipelines to produce highly realistic visualizations of generated outfits. Focus on accurate lighting, material shaders, and environmental integration.
- **Outfit Generation:** The system is designed to generate 4-5 distinct outfit combinations per prompt, ensuring variety and exploring different stylistic interpretations.
- **Quality Assurance:** A multi-stage QA process is implemented:
 - **Algorithmic Verification:** Automated checks for anatomical correctness, material consistency, and adherence to prompt parameters.
 - **Human Review:** A curated review by stylists and 3D artists to ensure aesthetic appeal, stylistic coherence, and overall quality.
 - **Feedback Loop:** Continuous refinement of generation algorithms based on QA feedback to improve output over time.

Summary

This technical report outlines the core modules of the Multi-Feature Generative Wardrobe & Style Prompt Engine. It details the parametric design of the physical architecture, the advanced fabric simulation and rendering capabilities of the digital engine, the biometric and stylistic analysis

methodologies, and the photorealistic image generation and quality assurance processes. The system is engineered for a minimalist, professional aesthetic, ensuring clear readability and printability.

I've built AuraCloset with @base44! <https://straight-aura-style-vault.base44.app>